

2_Partición_datos_externa

February 24, 2026

```
[1]: import pandas as pd
import numpy as np

#Leer la información del documento Excel original
dataFrame = pd.read_csv('../Material/Database.csv', sep=';')
print(dataFrame)
```

	ID	Class
0	OCTA1013661_19350130_1.png	Glaucoma
1	OCTA1153367_19320227_2.png	Glaucoma
2	OCTA1653454_19640601_2.png	Glaucoma
3	OCTADB5052_19500115_2.png	Glaucoma
4	OCTADB5551_19491106_1.png	Glaucoma
..	...	...
184	OCTADB7376_19590206_1.png	Healthy
185	OCTADB7376_19590206_2.png	Healthy
186	OCTADB7377_19620916_2.png	Healthy
187	OCTADB7398_19940401_1.png	Healthy
188	OCTADB7398_19940401_2.png	Healthy

[189 rows x 2 columns]

```
[3]: # Partición externa de los datos
from sklearn.model_selection import train_test_split, KFold

# train, test = train_test_split(dataFrame, test_size=0.2, shuffle=True,
    random_state=42) # hold-out

kf = KFold(n_splits=5, shuffle=True, random_state=42)
bolsas = kf.split(dataFrame)

for k, (train_fold, test_fold) in enumerate(bolsas):
    train = dataFrame.iloc[train_fold]
    test = dataFrame.iloc[test_fold]

lab_train, ocur_train = np.unique(train.Class, return_counts=True)
lab_test, ocur_test = np.unique(test.Class, return_counts=True)
```

```
print(' --- TRAIN --- \nGlaucoma: ', ocur_train[0], '\nHealthy: ',  
      ocur_train[1])  
print(' --- TEST --- \nGlaucoma: ', ocur_test[0], '\nHealthy: ', ocur_test[1])
```

```
--- TRAIN ---  
Glaucoma: 77  
Healthy: 75  
--- TEST ---  
Glaucoma: 16  
Healthy: 21
```

```
[4]: print(train)
```

	ID	Class
0	OCTA1013661_19350130_1.png	Glaucoma
2	OCTA1653454_19640601_2.png	Glaucoma
3	OCTADB5052_19500115_2.png	Glaucoma
4	OCTADB5551_19491106_1.png	Glaucoma
5	OCTADB6503_19290831_1.png	Glaucoma
..	...	...
181	OCTA6593993_19660107_2.png	Healthy
183	OCTADB5990_19431204_2.png	Healthy
184	OCTADB7376_19590206_1.png	Healthy
186	OCTADB7377_19620916_2.png	Healthy
187	OCTADB7398_19940401_1.png	Healthy

[152 rows x 2 columns]

```
[5]: # Aleatorizar los dataframes  
train = train.sample(frac=1)  
test = test.sample(frac=1)  
  
print(train)
```

	ID	Class
131	OCTADB7477_19380124_2.png	Glaucoma
83	OCTADB3311_19441225_2.png	Healthy
94	OCTADB7382_19750616_1.png	Healthy
114	OCTADB_19760911_1.png	Healthy
22	OCTADB7386_19400420_2.png	Glaucoma
..	...	...
0	OCTA1013661_19350130_1.png	Glaucoma
33	OCTADB7414_19660407_2.png	Glaucoma
25	OCTADB7399_19550917_2.png	Glaucoma
137	OCTA3956619_19651213_2.png	Healthy
61	OCTA1673572_19700825_1.png	Healthy

[152 rows x 2 columns]

```
[6]: # Guardado de los modelos
import os
if not os.path.exists('../partitions'):
    os.mkdir('../partitions')

train.to_csv('../partitions/train.csv', sep=';')
test.to_csv('../partitions/test.csv', sep=';')
```